

DESIGNING AND DEVELOPMENT OF AN INNOVATIVE EXTRUSION MACHINE FOR RECYCLABLE FILAMENT

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CONTENTS

Introduction	Briefing about project, Problem Statement, Aim and Objectives.
Literature Review	Previous work done by the researchers.
Methodology	The working and making of project.
Machine Hardware	These can be used in the template, and their size and color can be edited
Timeline	The management of time for project.

01

INTRODUCTION

BRIEFING ABOUT PROJECT

01

INTRODUCTION

PROBLEM STATEMENT

With the rise of 3D printing, plastic waste is becoming a growing issue, especially from failed prints, unused parts, and discarded filament. Currently, most filament is made from new, virgin plastics, which not only drives up costs but also adds to environmental pollution. Unfortunately, options for recycling these plastics back into high-quality filament are limited, as affordable and efficient machines for this purpose are scarce. This project seeks to fill that gap by designing an innovative extrusion machine that can transform waste materials from 3D printing into reusable filament, promoting sustainable recycling and helping to reduce both costs and plastic waste for circular economy.

INTRODUCTION

- **EXTRUSION MACHINE**

Equipment designed to melt and shape materials, here specifically for producing 3D printer filament from recycled plastics.

- **RECYCLABLE FILAMENT**

Filament made from waste plastic, suitable for reuse in 3D printing, supporting a circular economy.

- **3D PRINTING**

Additive manufacturing process that builds objects layer by layer using filaments, typically plastic, as material.

- **SUSTAINABLE MANUFACTURING**

Production process that minimizes environmental impact by reducing waste and reusing resources.

INTRODUCTION

- **TEMPERATURE CONTROL**

Precise regulation of heat within the extrusion machine is to ensure consistent melting and filament quality.

- **FILAMENT CONSISTENCY**

Uniformity in filament thickness and strength, is essential for reliable 3D printing.

- **ECO-FRIENDLY PRODUCTION**

Manufacturing approach that reduces carbon footprint and waste, aligning with environmental sustainability goals.

- **MATERIAL OPTIMIZATION**

Process of selecting and treating recycled plastics for maximum performance and quality in filament output

INTRODUCTION

AIM

To design and develop an innovative extrusion machine for recyclable filaments.

OBJECTIVES

- To design a low-cost extrusion machine that provides an affordable solution without compromising on quality or efficiency.
- To develop an innovative and portable design for the machine, ensuring ease of use and mobility for various applications.
- To recycle waste filament generated from 3D printing, allowing for the reuse of materials and promoting sustainability in the 3D printing process.

INTRODUCTION

SDGs

GOAL ID	SUSTAINABLE DEVELOPMENT GOAL	DESCRIPTION
Goal 9	Industry, Innovation, and Infrastructure	The design of an extrusion machine for recyclable filament promotes innovative manufacturing techniques that enhance the efficiency of the 3D printing industry by using sustainable, recyclable materials. The project helps advance infrastructure in the additive manufacturing industry by reducing reliance on non-recycled filaments and contributing to efficient, cost-effective material production.
Goal 12	Responsible Consumption and Production	The development of this extrusion machine for recyclable filament production directly aligns with SDG 12 by promoting responsible consumption and production. By focusing on recycling plastics and converting waste

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LITERATURE REVIEW

PREVIOUS WORK DONE BY RESEARCHERS

PAPER #	TITLE	AUTHOR	PUBLISHED YEAR	REMARKS
1	Design And Fabrication Of Extrusion Machine For Recycling Plastics	S Kumar , R Sooraj, MVV Kumar	2021	• The extruder is designed to produce 3D printer filament with a diameter range of 2-2.5 mm. • The screw extrusion process is used to achieve a fixed cross-section for the filament. • PET pellets are heated to temperatures between 230°C and 250°C and extruded through a nozzle. • The process ensures a continuous filament free of air bubbles and roughness, meeting the specified quality requirements
2	Design and development of 3D printer filament extruder	M Aris, S Mat , MS Sam, FR Ramli , MR Alkahari , SIA Kudus	2020	• The project aims to recycle 3D printing waste into usable filament using an extruder machine, focusing on recycled Acrylonitrile Butadiene Styrene (ABS). • The recycled filament is utilized in an open-source 3D printer to create a specimen. • Mechanical properties, including tensile strength, tensile strain, maximum load, and Young's modulus, are compared between recycled and pure ABS. • Results indicate that the original ABS specimen can withstand a maximum load of 724N, while the recycled ABS specimen can handle 713N.

PAPER #	TITLE	AUTHOR	PUBLISHED YEAR	REMARKS
3	Developing Filament Extruder and Characterization of Recycled High-Density Polyethylene for 3D Printing Filament Material	M Shiferaw, H Gebremedhe , A Mengistie	2023	• The study developed a desktop-sized filament extruder machine to create 1.75 mm diameter recycled HDPE filament for 3D printing, achieving a smooth, round shape suitable for prototyping. • Mechanical and thermal tests showed that the recycled HDPE filament had an ultimate tensile strength of 19.02 MPa, comparable to virgin HDPE, along with high thermal stability (degradation onset at 430°C). • FTIR analysis confirmed that the recycled HDPE retained the same functional groups as virgin HDPE, and it also displayed excellent water-repellent properties. • The study suggests that recycled HDPE filament could be a sustainable alternative for 3D printing, but further research is needed to enhance filament quality, optimize extrusion parameters, and improve product performance.
4	Development of filament extruder using recycled plastic material for 3D printer	Jen Aldwayne M., Banaag	2016	• Developed an extruder machine to recycle waste plastic bottles into 1.00-1.75 mm diameter 3D printer filament. • Controlled heating (220-260°C) and Arduino automation ensured consistent output. • Motor speed and cooling fan were key to achieving stable filament for successful 3D printing

LITERATURE REVIEW

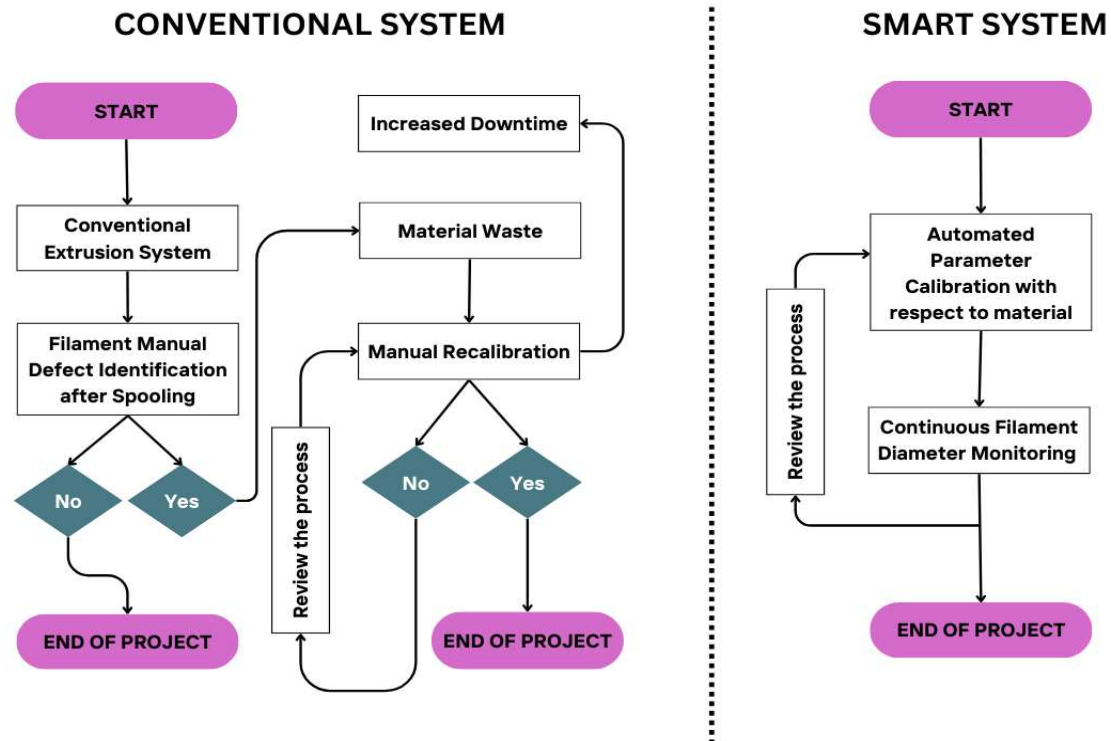
PAPER #	TITLE	AUTHOR	PUBLISHED YEAR	REMARKS
5	Temperature-induced pressure gradient in the clearance between screw flight and barrel of a single screw extruder	Winter, Horst Henning.	1980	- Examines temperature-induced pressure gradients in drag flow with a viscosity gradient due to temperature changes. - Focuses on clearance flow in single screw extruders, where a polymer layer prevents metal-to-metal contact. - Concludes that failure of this layer, leading to dry friction, is caused by temperature-induced pressure flow.
6	Design and Manufacture of the Cooling and Collection System of a Single Screw Extruder	Opeyemi Osideko, Favour Eboh, and Modupe Onitiri	5 April 2022	- Focuses on designing and constructing a lightweight, cost-effective filament collection system to improve cooling efficiency in a single screw extruder. - Implements a filament winding machine with a circumferential winding pattern, integrating a motor, gear system, and additional cooling fan, leading to a 35% increase in cooling rate. - Highlights the benefits of an automated collection system in reducing filament waste, contamination, and enhancing productivity.

03

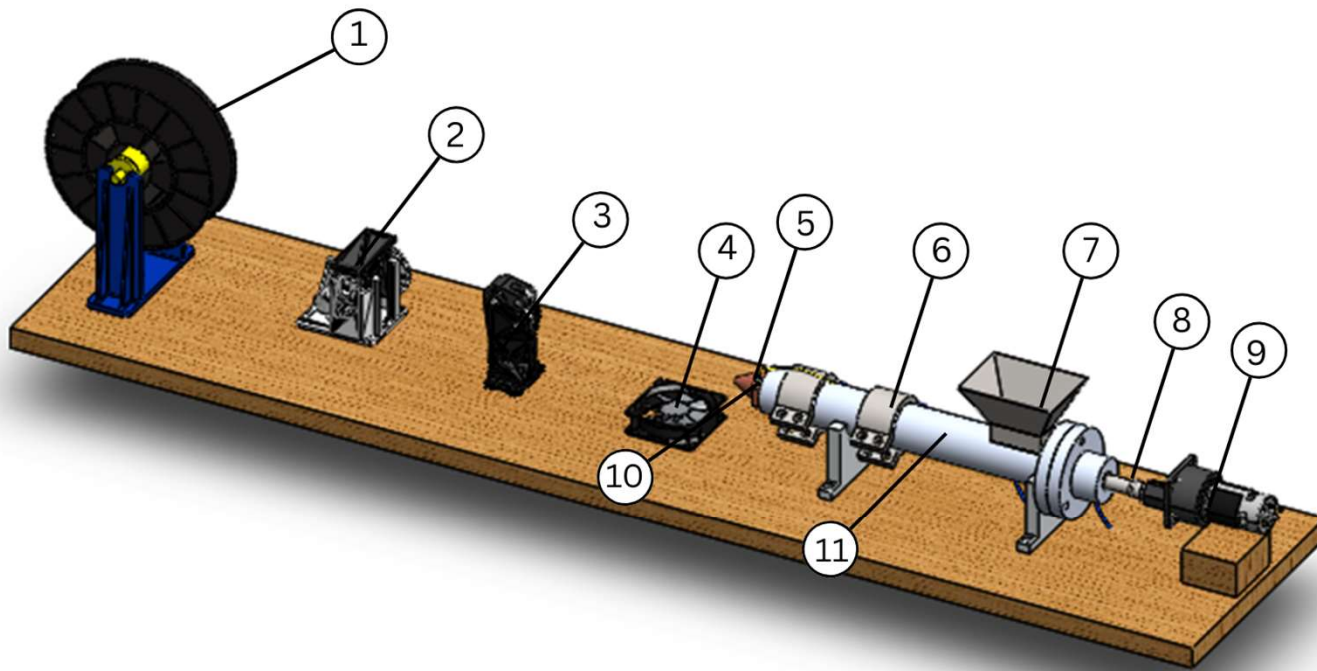
METHODOLOGY

MAKING AND WORKING OF PROJECT

CONVENTIONAL VS SMART EXTRUSION SYSTEM

*Figure 1: Conventional vs Smart Extrusion System*

3D MODEL



1. Winding Spool
2. Puller Assembly
3. Filament Diameter Sensor
4. Cooling Fan
5. Thermocouple
6. Heaters
7. Funnel
8. Coupling
9. DC Motor
10. Nozzle
11. Barrel

METHODOLOGY

COST ESTIMATION

Sr. No.	Product	Price (PKR)
1	Cooling Fan	840
2	Thermocouple	842
3	Bread Board	250
4	LCD Display	400
5	Jumping Wires	360
6	Relay 5 v	130
7	Auger drill bit	3706
8	Barrel	325
9	Clamp	30
10	Nozzle	275
11	Amplifier MAX 6675	350
12	Motor	4000
13	Motor Adapter Wire	450
14	Commercial Board	1300
15	Bearing	750
16	Machining	5500
17	Heater	3200
18	8 Bearing	800
19	Puller Motor	600
20	Magnet	70
21	I2c	250
22	Hall Effect	150
23	L298 Controllor	450
Total		25478

MAJOR DESIGN CONSIDERATIONS

SR. NO.	PART NAME	DESIGN CONSIDERATION	ABBREVIATIONS	DIMENSIONS
1	Barrel	Outer Diameter	B_o	32 mm
		Inner Diameter	B_i	28 mm
		Barrel Length	L	460 mm
2	Nozzel	Internal Diameter	N_i	32 mm
		External Diameter	N_o	36 mm
		Hole	N_r	1.7 mm
3	Heater	Radius	H_r	36 mm
		Length	H_l	100 mm
4	Extrusion Bit	Outer Diameter	E_o	28 mm
		Shaft Diameter	E_s	10 mm

MAIN CALCULATIONS

VOLUMETRIC FLOW RATE

$$Q = A \cdot v = 4\pi d^2 \cdot v$$

d = diameter of extruded filament [m]

v = linear extrusion speed [m/s]

Q = volumetric flow rate [m³/s]

$$Q = 3.14(2)^2(10)/4$$

$$Q = 31.4 \text{ m}^3/\text{s}$$

SHEAR RATE ($\dot{\gamma}$) AT THE NOZZLE

$$\dot{\gamma} = 4Q/\pi R^3$$

$\dot{\gamma}$ = Shear rate [s⁻¹]

Q = Volumetric flow rate [m³/s]

R = Radius of nozzle [m]

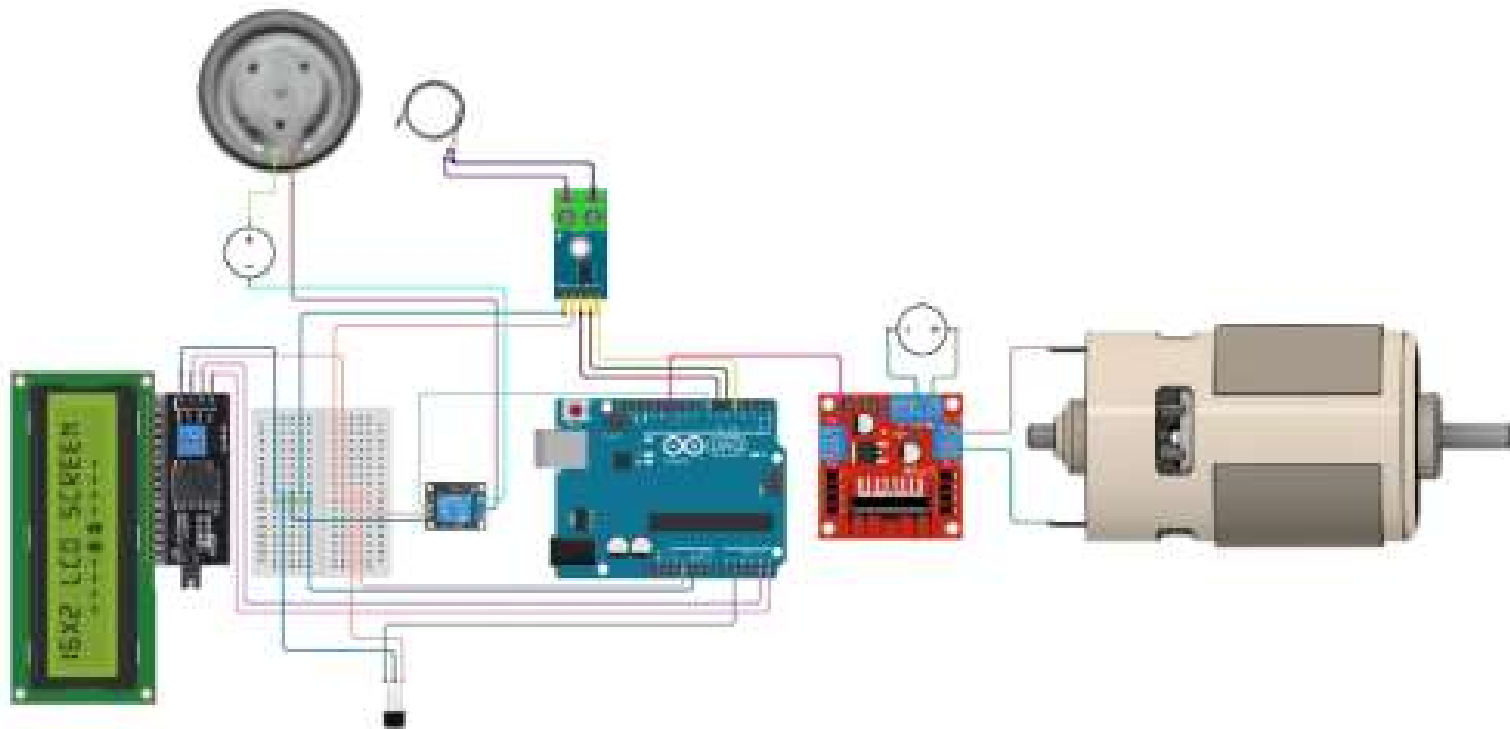
$$\dot{\gamma} = 4(31.4) / (3.14) (1)^3$$

$$\dot{\gamma} = 40 \text{ s}^{-1}$$

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METHODOLOGY

CIRCUIT DIAGRAM



METHODOLOGY

ARDUINO CODE

```
// Include necessary libraries
#include <Wire.h> // Required for I2C communication
#include <LiquidCrystal_I2C.h> // For the I2C LCD
#include <max6675.h> // For the thermocouple

// --- I2C LCD Setup ---
// Set the LCD address (common addresses are 0x27 or 0x3F)
// If unsure, use an I2C Scanner sketch to find the correct address
LiquidCrystal_I2C lcd(0x27, 16, 2); // Address 0x27, 16 columns, 2 rows

// --- Thermocouple Pins ---
int thermoDO = 4;
int thermoCS = 5;
int thermoCLK = 6;
MAX6675 thermocouple(thermoCLK, thermoCS, thermoDO);

// --- Heater Relay ---
int heaterRelay = 7;
```

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METHODOLOGY

ARDUINO CODE

```
// --- L298 Motor Pins ---
int enA = 9; int in1 = 8; int in2 = 10; // Screw Motor
int enB = 11; int in3 = 12; int in4 = 13; // Puller Motor

// --- Filament Sensor ---
#define NUMTEMPS 4
#define smooth 50
#define sensIn A1

// Lookup Table for filament diameter calibration
// * IMPORTANT: Check if these values match your sensor readings! *
static float lut3[NUMTEMPS][2] = {
  {494, 0},
  {495.1, 0.5},
  {498.0, 1},
  {512.8, 2.5}
};

float sensbuf = lut3[0][0]; // Initialize smoothed buffer
```

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METHODOLOGY

ARDUINO CODE

```
// --- Target Control Variables ---  
float targetTemp = 190;  
float targetDiameter = 1.75;  
  
int baseSpeed = 150;  
int pullerSpeed = baseSpeed;  
  
// --- Variables for LCD display timing & state ---  
unsigned long lastLcdUpdateTime = 0;  
// * Adjust LCD Update Speed Here (milliseconds) *  
const unsigned long lcdUpdateInterval = 1500; // Update LCD every 1.5 seconds (change as needed)  
  
String lastStatusMessage = "";  
bool showStatusOnLcd = false; // Flag to alternate LCD line 2  
  
void setup() {  
  Serial.begin(9600);  
  // Serial.println("Smart Extrusion System Initializing..."); // Optional: Keep or remove  
  // Serial.println("Initialization Complete."); // Optional: Keep or remove
```

METHODOLOGY

ARDUINO CODE

```
// Initialize LCD
lcd.init(); // Initialize the LCD
lcd.backlight(); // Turn on the backlight
lcd.setCursor(0, 0);
lcd.print("Initializing...");

// Initialize Pins
pinMode(heaterRelay, OUTPUT);
pinMode(in1, OUTPUT); pinMode(in2, OUTPUT);
pinMode(in3, OUTPUT); pinMode(in4, OUTPUT);

// Set motors to forward initially
digitalWrite(in1, HIGH); digitalWrite(in2, LOW);
digitalWrite(in3, HIGH); digitalWrite(in4, LOW);

delay(1000); // Wait a bit for sensors to stabilize
lcd.clear();
}
```

METHODOLOGY

ARDUINO CODE

```
void loop() {  
  // Read current temperature  
  float currentTemp = thermocouple.readCelsius();  
  
  // Heater ON/OFF Control (Bang-Bang)  
  if (currentTemp < targetTemp - 5) {  
    digitalWrite(heaterRelay, HIGH);  
  } else if (currentTemp > targetTemp + 5) {  
    digitalWrite(heaterRelay, LOW);  
  }  
  bool heaterState = digitalRead(heaterRelay);  
  
  // --- Read filament diameter ---  
  int rawSensor = analogRead(sensIn);  
  sensbuf += (rawSensor - sensbuf) / smooth;  
  float diameter = lookup(sensbuf, lut3);  
  
  // --- Adaptive Adjustment Logic ---  
  float error = diameter - targetDiameter;  
  String currentStatus = ""; // Status for this loop iteration  
  bool printStatusLine = false; // Flag to print status on a new line
```

METHODOLOGY

ARDUINO CODE

```
if (diameter > 0 && error > 0.05) { // Only adjust if diameter reading is valid
  // Too thick -> pull faster
  pullerSpeed += 10;
  pullerSpeed = constrain(pullerSpeed, 100, 255);
  currentStatus = "Filament too thick → Increasing puller speed";
  printStatusLine = true;
} else if (diameter > 0 && error < -0.05) { // Only adjust if diameter reading is valid
  // Too thin -> increase temp (original logic)
} else if (diameter > 0 && error < -0.05) {
  // Too thin → pull slower
  pullerSpeed -= 10;
  pullerSpeed = constrain(pullerSpeed, 100, 255); // You can allow down to 80 or lower if needed
  currentStatus = "Filament too thin → Decreasing puller speed";
  printStatusLine = true;
} else if (diameter <= 0) {
  currentStatus = "Sensor Error?"; // Keep this for LCD
  // Don't print a specific status line for sensor error in the requested format
} else {
  currentStatus = "Diameter OK"; // Keep this for LCD
  // Don't print a status line if diameter is OK
}
```

METHODOLOGY

ARDUINO CODE

```
// Update the status message for LCD if it changed
if (currentStatus != lastStatusMessage) {
    lastStatusMessage = currentStatus;
}

// Set Motor Speeds
analogWrite(enA, baseSpeed);    // Screw motor
analogWrite(enB, pullerSpeed);  // Puller motor

// --- Serial Monitoring (Formatted as requested) ---
Serial.print("Temp: "); Serial.print(currentTemp, 2); // Use 2 decimal places like original example
Serial.print(" °C | Heater: "); Serial.print(heaterState ? "ON" : "OFF");
Serial.print(" | Diameter: "); Serial.print(diameter, 3);
Serial.print(" mm | Puller Speed: "); Serial.println(pullerSpeed);

// Print status message on a new line if needed
if (printStatusLine) {
    Serial.println(currentStatus);
}
```

03

METHODOLOGY

ARDUINO CODE

```
// --- LCD Update Logic (Unchanged) ---
unsigned long currentTime = millis();
if (currentTime - lastLcdUpdateTime >= lcdUpdateInterval) {
  lastLcdUpdateTime = currentTime;
  showStatusOnLcd = !showStatusOnLcd; // Toggle flag each update

  // Line 0: Temperature and Heater Status (Always shown)
  lcd.setCursor(0, 0);
  String line0 = "T:" + String(currentTemp, 1) + " H:" + (heaterState ? "ON " : "OFF");
  lcd.print(line0);
  // Pad with spaces to clear rest of line
  for (int i = line0.length(); i < 16; i++) { lcd.print(" "); }

  // Line 1: Alternate between Diameter/Puller and Status Message
  lcd.setCursor(0, 1);
  String line1 = "";
```

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METHODOLOGY

ARDUINO CODE

```
if (showStatusOnLcd && lastStatusMessage != "") {  
    // Display the status message (use the internal status, e.g., "Thin->Inc Temp")  
    if (lastStatusMessage == "Filament too thick → Increasing puller speed") line1 = "Thick->Pull Faster";  
    else if (lastStatusMessage == "Filament too thin → Increasing temp slightly") line1 = "Thin->Inc Temp";  
    else line1 = lastStatusMessage; // Display other statuses like "Sensor Error?" or "Diameter OK"  
} else {  
    // Display Diameter and Puller Speed  
    line1 = "D:" + String(diameter, 2) + " P:" + String(pullerSpeed);  
}  
lcd.print(line1);  
// Pad with spaces to clear rest of line  
for (int i = line1.length(); i < 16; i++) { lcd.print(" "); }  
}  
  
// Short delay - adjust if needed, but LCD updates control the main timing  
delay(1000);  
}
```

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METHODOLOGY

ARDUINO CODE

```
// --- Lookup Table Interpolation Function (Unchanged) ---
float lookup(float inval, float lut[][2]) {
    byte i;
    // Check if below the first entry
    if (inval <= lut[0][0]) {
        return lut[0][1]; // Return the diameter associated with the lowest raw value (0 in this case)
    }
    // Check if above the last entry
    if (inval >= lut[NUMTEMPS - 1][0]) {
        return lut[NUMTEMPS - 1][1]; // Return max diameter
    }
    // Find the interval and interpolate
    for (i = 1; i < NUMTEMPS; i++) {
        if (lut[i][0] >= inval) { // Find the first raw value in the table >= input value
            // Interpolate within the interval [i-1, i]
            float val1 = lut[i-1][0]; // Lower raw value
            float val2 = lut[i][0];   // Upper raw value
            float out1 = lut[i-1][1]; // Lower diameter
            float out2 = lut[i][1];   // Upper diameter
```

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METHODOLOGY

ARDUINO CODE

```
// Perform linear interpolation
return out1 + (inval - val1) * (out2 - out1) / (val2 - val1);
}
}
// Should not be reached due to checks above, but return max as fallback
return lut[NUMTEMPS - 1][1];
}
```

04

HARDWARE

ACTUAL WORKING MACHINE MADE

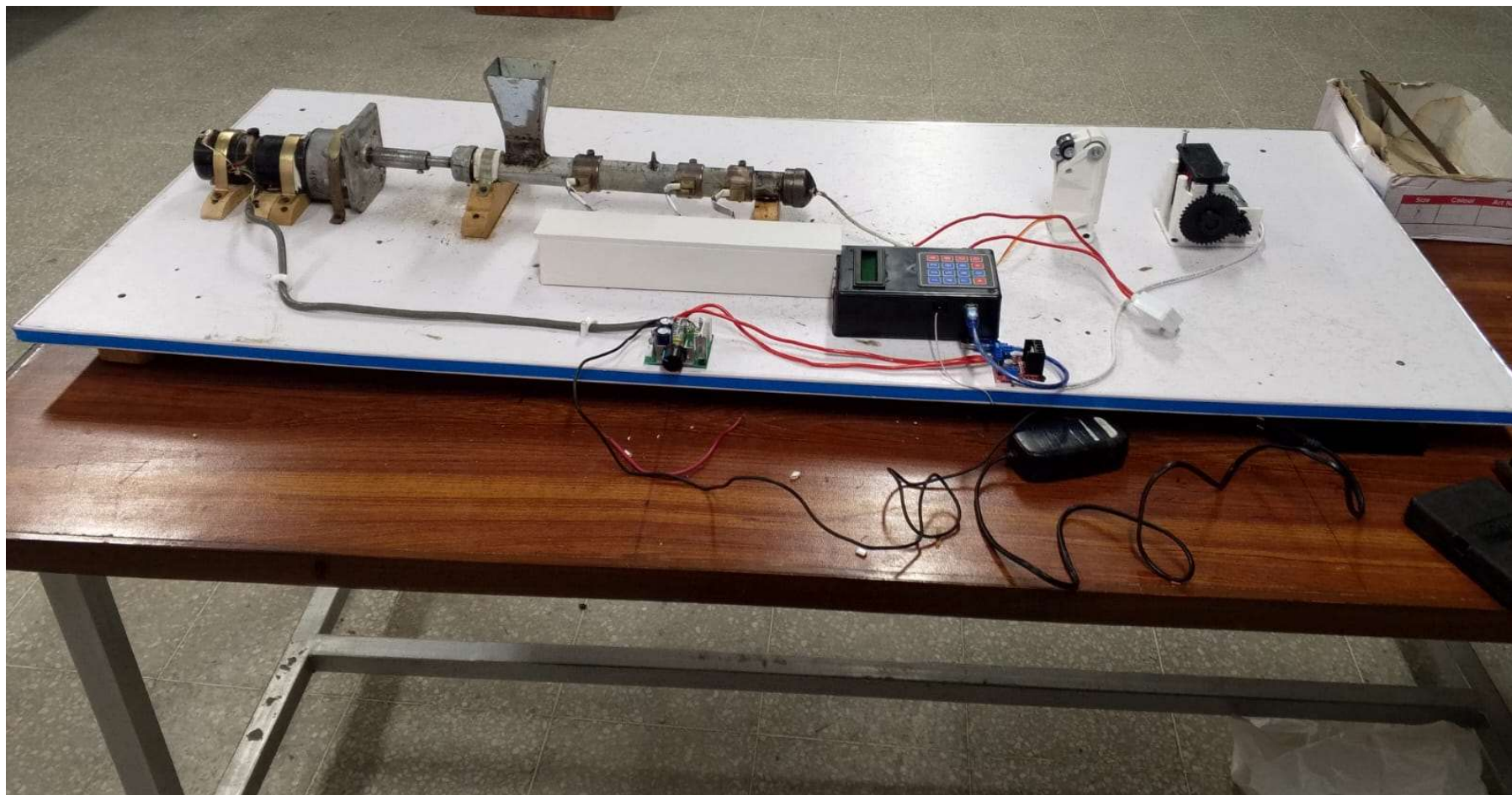
04

HARDWARE



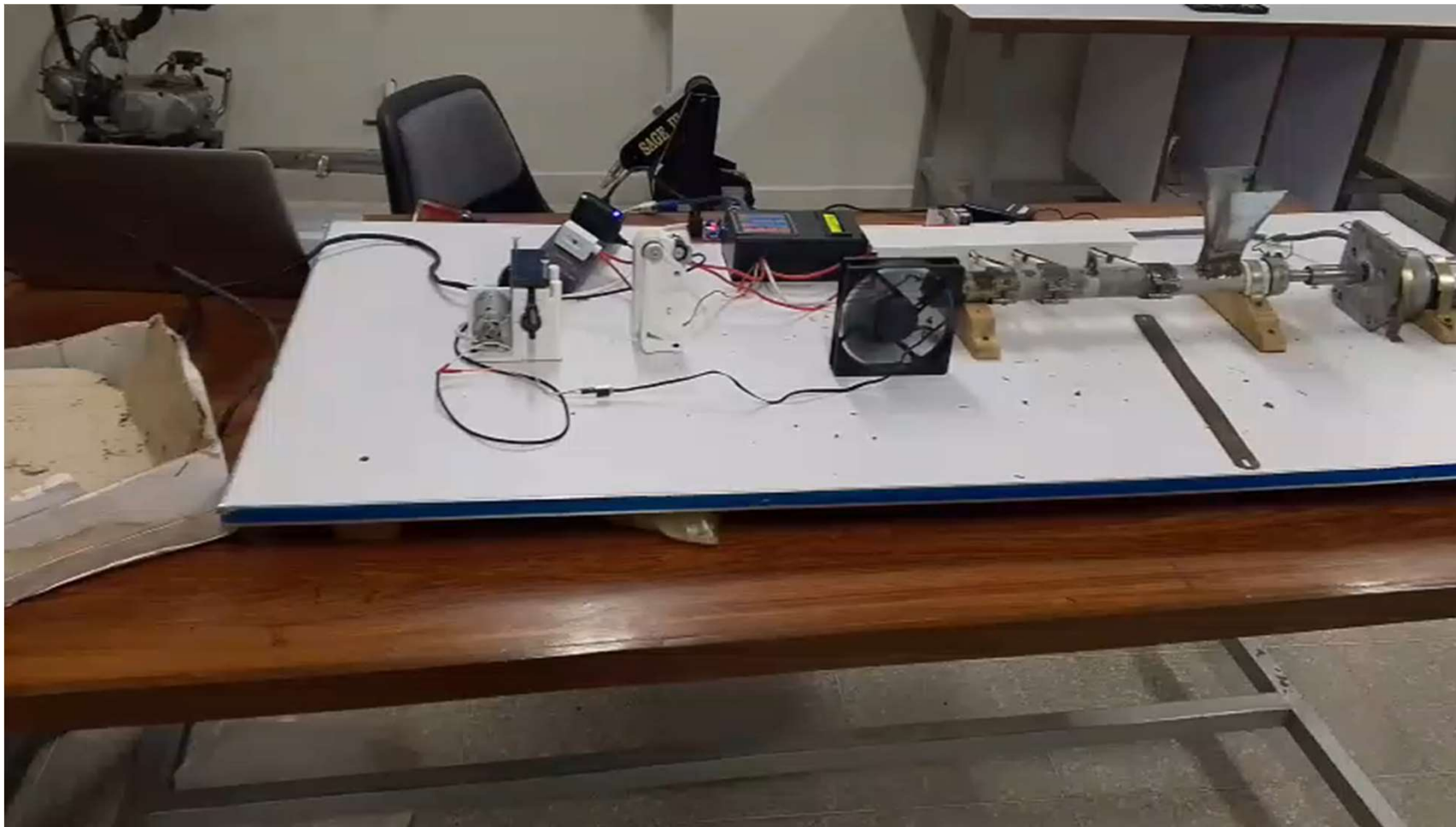
04

HARDWARE



04

HARDWARE



04

HARDWARE

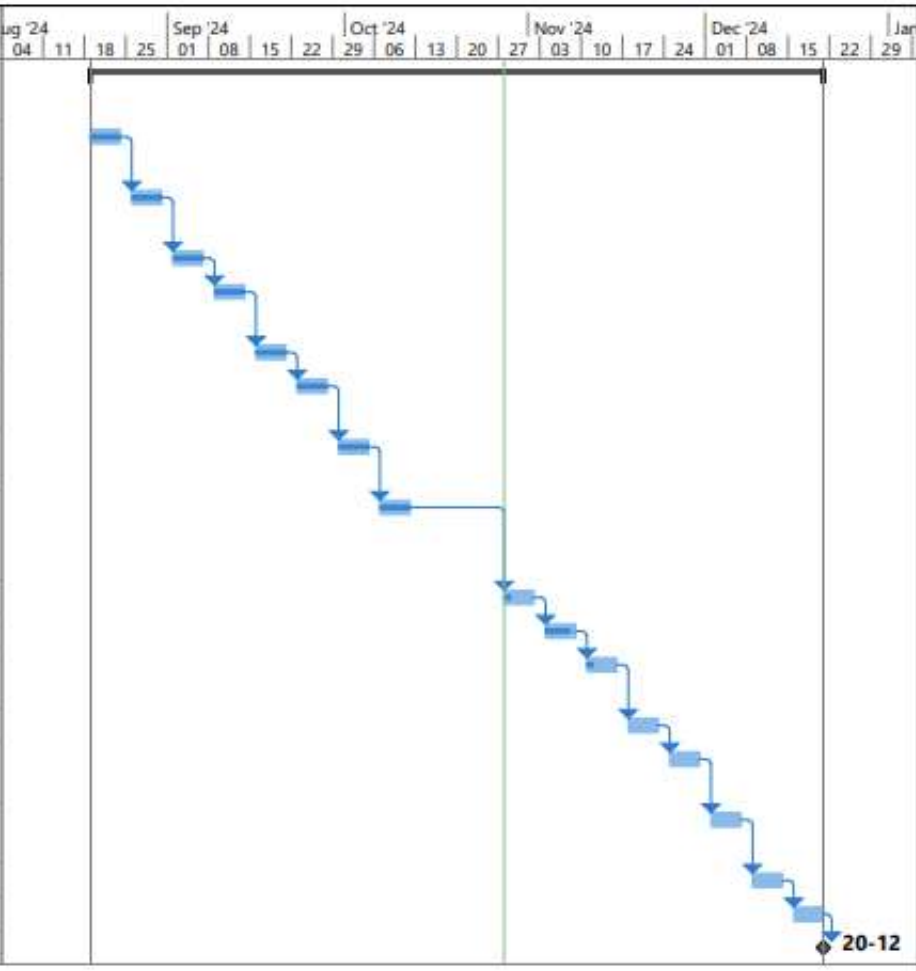


05

TIMELINE

TIME MANAGEMENT FOR PROJECT

ID	Task Name	Duration	Start	Finish	Status	Predecessors	Aug '24				Sep '24				Oct '24				Nov '24				Dec '24				Jan
							04	11	18	25	01	08	15	22	29	06	13	20	27	03	10	17	24	01	08	15	
1	Project Phase 1 (7th Semester)	90 days	Mon 19-08-24	Fri 20-12-24	On Schedule																						
2	Week 1 (FYP Allotment)	5 days	Mon 19-08-24	Fri 23-08-24	Complete																						
3	Week 2 (Introduction)	5 days	Mon 26-08-24	Fri 30-08-24	Complete	2																					
4	Week 3 (Research)	5 days	Mon 02-09-24	Fri 06-09-24	Complete	3																					
5	Week 4 (Research & 3D Modeling)	5 days	Mon 09-09-24	Fri 13-09-24	Complete	4																					
6	Week 5 (MRP)	5 days	Mon 16-09-24	Fri 20-09-24	Complete	5																					
7	Week 6 (Material Order online and offline)	5 days	Mon 23-09-24	Fri 27-09-24	Complete	6																					
8	Week 7 (Started work on Reporting)	5 days	Mon 30-09-24	Fri 04-10-24	Complete	7																					
9	Week 8 (Coding of Arduino and Designed Electric Connections)	5 days	Mon 07-10-24	Fri 11-10-24	Complete	8																					
10	Week 9 (Evaluation)	5 days	Mon 28-10-24	Fri 01-11-24	On Schedule	9																					
11	Week 10 (Assembly)	5 days	Mon 04-11-24	Fri 08-11-24	Future Task	10																					
12	Week 11 (Assembly and Experimentation)	5 days	Mon 11-11-24	Fri 15-11-24	Future Task	11																					
13	Week 12 (Testing)	5 days	Mon 18-11-24	Fri 22-11-24	Future Task	12																					
14	Week 13 (Calculations for Design of Experiment)	5 days	Mon 25-11-24	Fri 29-11-24	Future Task	13																					
15	Week 14 (Analysis through MINITAB)	5 days	Mon 02-12-24	Fri 06-12-24	Future Task	14																					
16	Week 15 (Reporting)	5 days	Mon 09-12-24	Fri 13-12-24	Future Task	15																					
17	Week 16 (Lab Mid Exams)	5 days	Mon 16-12-24	Fri 20-12-24	Future Task	16																					
18	Milestone Phase 1	0 days	Fri 20-12-24	Fri 20-12-24	Future Task	17																					



06

RESULTS & ANALYSIS

DIAMETER SENSOR VALIDATION



06

RESULTS & ANALYSIS

PULLER CONTROL LOGIC TESTING

- When the filament is thick counted by diameter sensor then signal is sent to L298 through Arduino to increase the puller speed.
- When the filament is thin counted by diameter sensor then signal is sent to L298 through Arduino to decrease the puller speed



RESULTS & ANALYSIS

CFD ANALYSIS

- **Objective:** To understand material flow inside the extrusion barrel using CFD analysis.
- **Challenge:** Internal flow is unobservable due to barrel confinement and material opacity.
- **Solution:** CFD provides a non-invasive method to visualize and analyze flow dynamics.
- **Benefit:** Helps verify design validity and mechanical soundness of the extrusion system.
- **Focus:** Simulated thermal and flow behavior of molten PLA through the screw and nozzle.

06

RESULTS & ANALYSIS

GEOMETRY PREPARATION AND MESHING

Tool Used: ANSYS Meshing with CFD physics preference and Fluent solver

Mesh Strategy: Linear element order selected for faster computation under laminar flow

Element Size: Default size 25.102 mm with defeaturing to simplify minor geometry

Mesh Refinement: Curvature and proximity capturing enabled for complex shapes

Key Parameters:

- Curvature min: 2.0 mm
- Proximity: 1.0 mm
- Defeature size: 0.12551 mm
- Mesh growth rate: 1.2 (for stability)

Element Details:

- Max element size: 50.204 mm
- Avg surface area: 1262.1 mm²
- Min edge length: 0.88942 mm

Focus Areas: Enhanced resolution near screw surfaces and nozzle for accurate flow prediction

Efficiency: Balanced mesh quality with low computational load

06

RESULTS & ANALYSIS

DETAILS OF MESH

Details of "Mesh" ▾ 🔍 🗑️ ✕	
[-] Display	
Display Style	Use Geometry Setting
[-] Defaults	
Physics Preference	CFD
Solver Preference	Fluent
Element Order	Linear
☐ Element Size	Default (25.102 mm)
Export Format	Standard
Export Preview Surface Mesh	No
[-] Sizing	
Use Adaptive Sizing	No
☐ Growth Rate	Default (1.2)
☐ Max Size	Default (50.204 mm)
Mesh Defeaturing	Yes
☐ Defeature Size	Default (0.12551 mm)
Capture Curvature	Yes
☐ Curvature Min Size	2.0 mm
☐ Curvature Normal Angle	Default (18.0°)
Capture Proximity	Yes
☐ Proximity Min Size	1.0 mm
☐ Num Cells Across Gap	Default (3)
Proximity Size Function Sou...	Faces and Edges
Bounding Box Diagonal	502.04 mm
Average Surface Area	1262.1 mm ²
Minimum Edge Length	0.88942 mm
[+] Quality	
[+] Inflation	
[+] Advanced	
[+] Statistics	

RESULTS & ANALYSIS

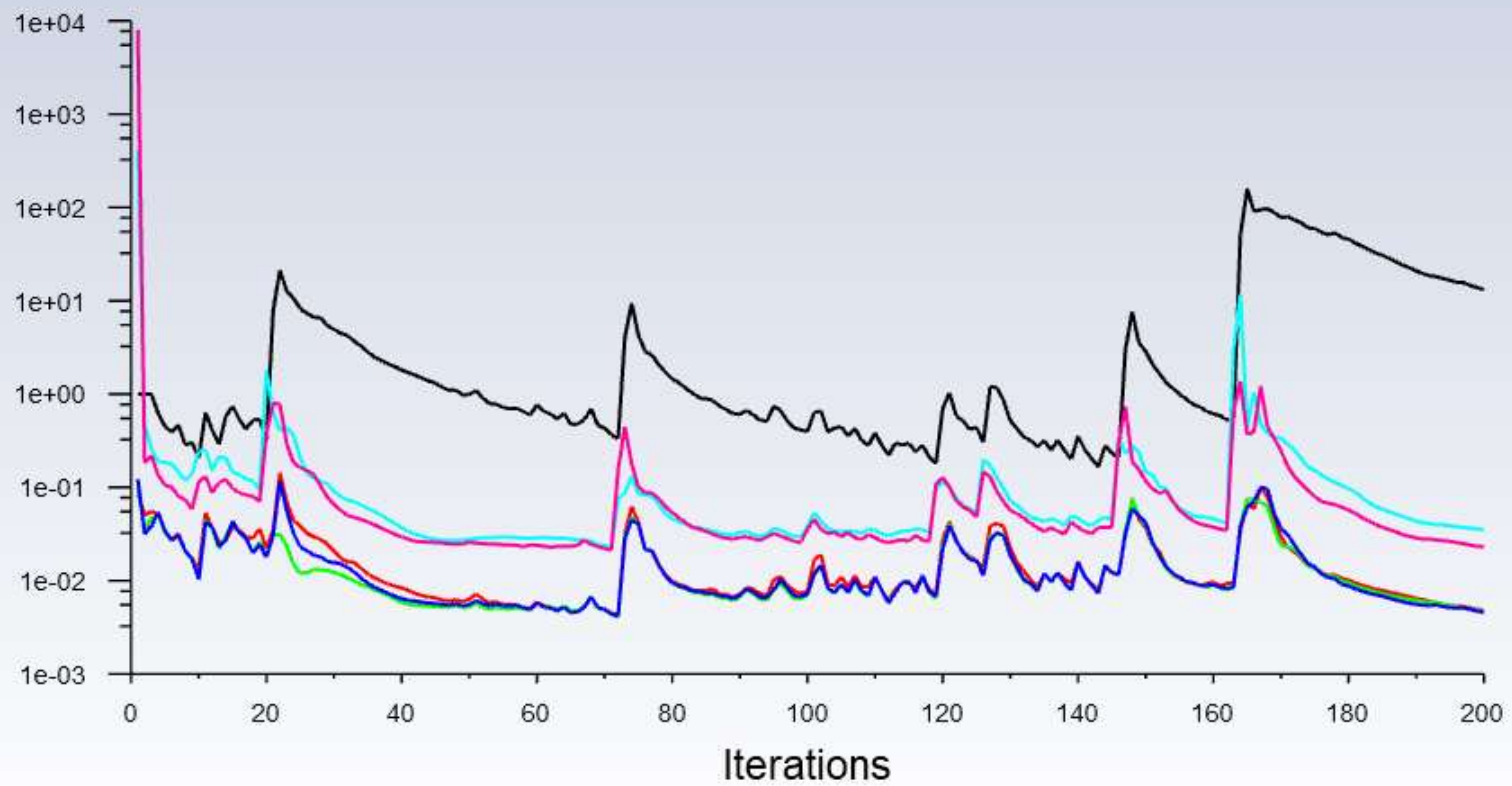
BOUNDARY CONDITION & SOLVER SETUP

- **Boundary Conditions:** Set to reflect realistic operating environment
- **Material Model:** Molten PLA treated as incompressible, non-Newtonian, high-viscosity fluid
- **Flow Type:** Laminar flow assumption due to low Reynolds number
- **Inlet Condition:** Mass flow inlet based on experimental extrusion rate
- **Outlet Condition:** Pressure outlet set to atmospheric pressure (open-end discharge)
- **Thermal Setup:** Barrel walls maintained at constant 175 °C (heater environment)
- **Wall Condition:** No-slip condition assumed for full fluid-wall adherence
- **Solver Used:** Steady-state, pressure-based solver suitable for high-viscosity laminar flow.

06

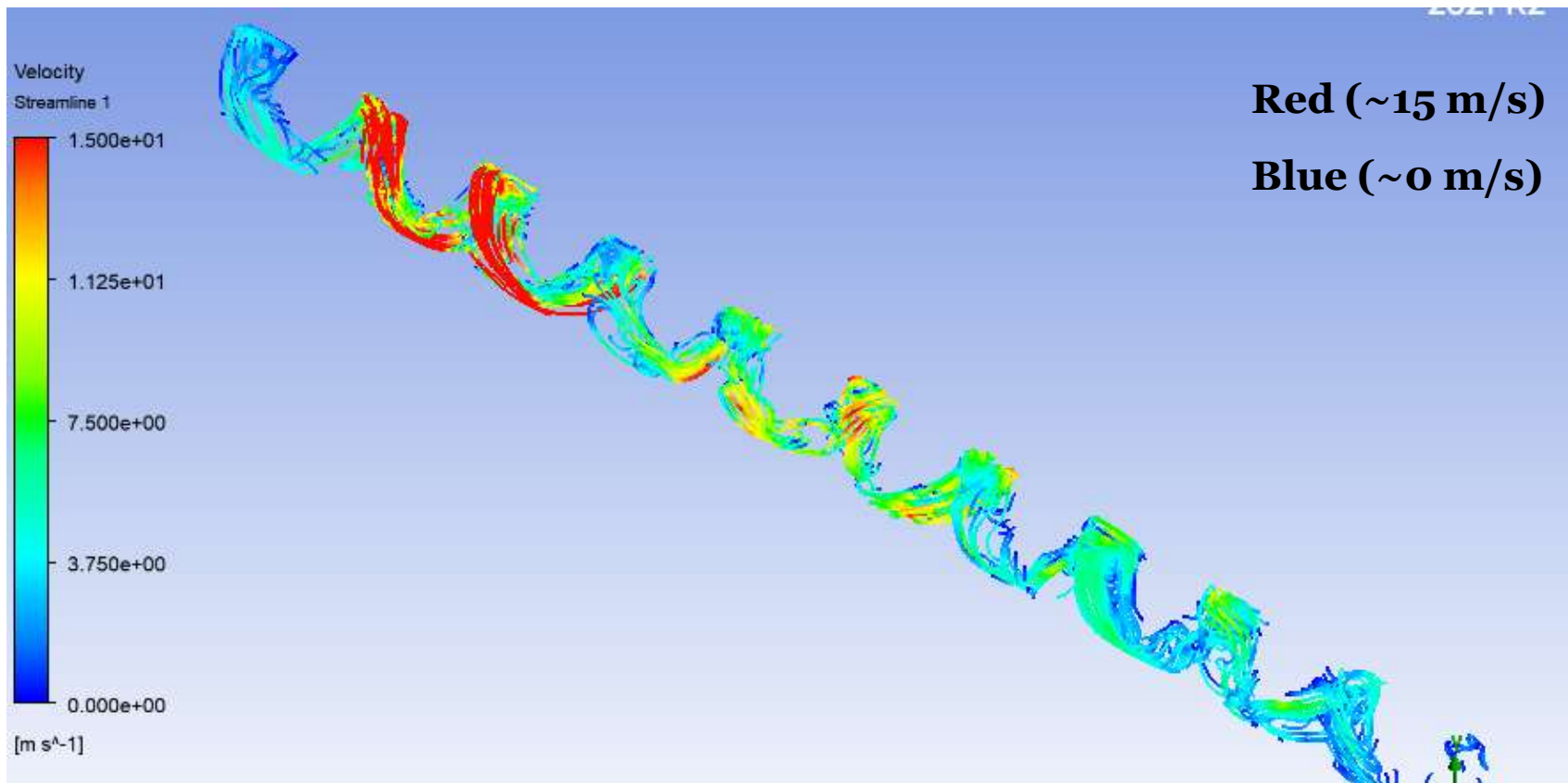
RESULTS & ANALYSIS

ITERATIONS VS VELOCITY



RESULTS & ANALYSIS

CFD FLOW ANALYSIS



CONCLUSIONS

- The system integrates key components with an Arduino Uno-based control unit.
- Sensor-based automation was achieved using real-time feedback from a K-type thermocouple and diameter sensor, enabling thermal regulation and filament sizing through closed-loop control.
- The CAD-based mechanical design ensured compactness and ease of fabrication using accessible materials.
- CFD and thermal-structural simulations validated the system's flow behaviour, heat distribution, and mechanical stability, reducing reliance on physical trials.
- The project confirms the feasibility of a self-regulating, sustainable filament extrusion unit for eco-friendly 3D printing applications.

06

ENDING

THANK

YOU